

SYSTEMS PROJECT MANAGEMENT

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



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Information Systems

- An **Information System** is a **system** that collects, processes, stores, and disseminates information.
- Information systems are used to support and manage business operations.
- Information systems are used to automate and optimize business processes.
- Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.

→ first Project



Information Systems

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Processes

computer
scientist



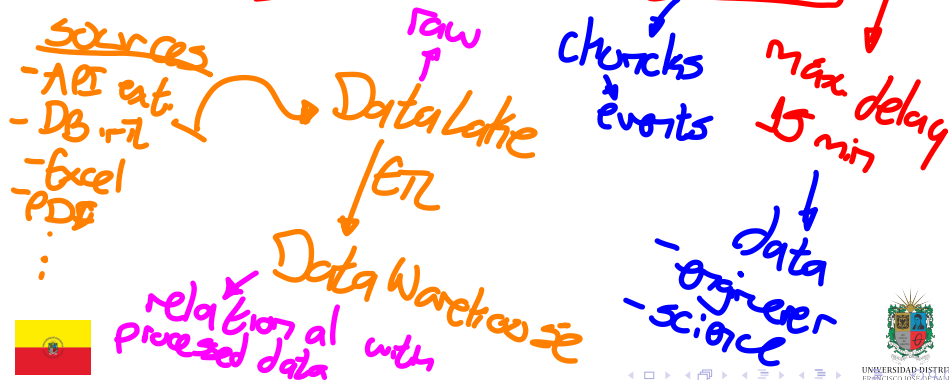
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Data Systems

- A **Data System** is a **system** that **collects**, **processes**, **stores**, and **retrieves data**.
- **Data systems** are used to **store** and **analyze** data.
- Examples of **data systems** include databases, data warehouses, data lakes, data marts, data cubes and data streams.



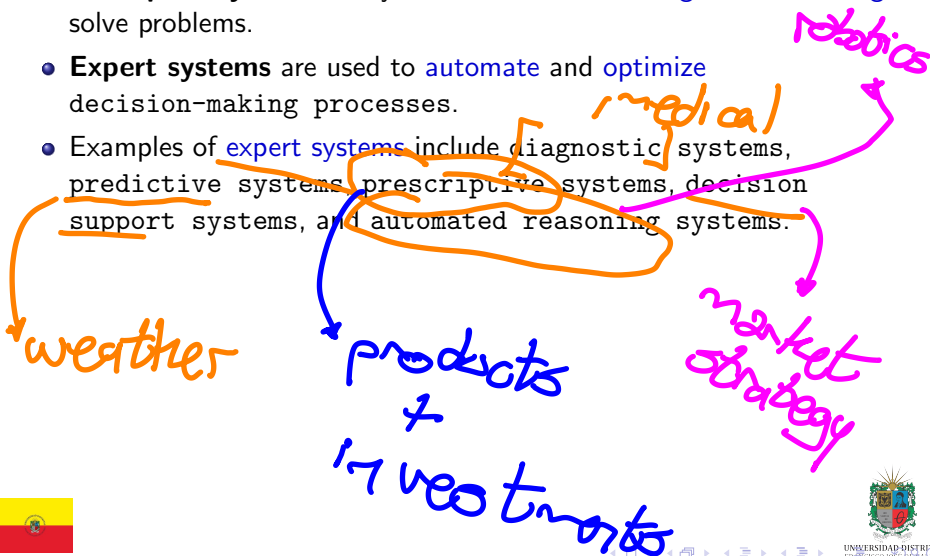
Expert Systems

- An **Expert System** is a system that uses **knowledge** and **reasoning** to solve problems. *90's*
- **Expert systems** are used to **automate** and **optimize** **decision-making processes**. *Partial*
AGI
- Examples of **expert systems** include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems.



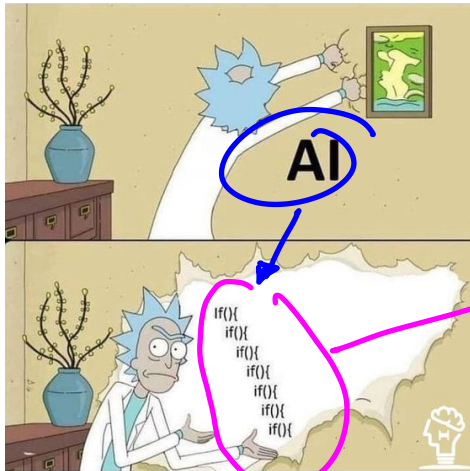
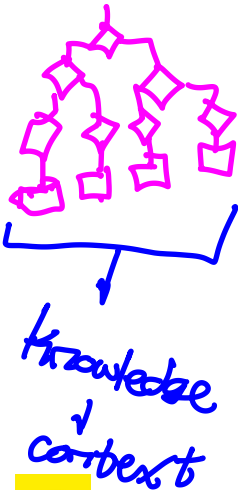
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- **Expert systems** are used to **automate** and **optimize** decision-making processes.
- Examples of **expert systems** include **diagnostic systems**, **predictive systems**, **prescriptive systems**, **decision support systems**, and **automated reasoning systems**.



Expert Systems as Classical Artificial Intelligence

Here there is a great example of a **diagnostic system**.



decision
tree



Risks and Failures in Information

- **Information systems** are subject to **risks** and failures that can impact business operations. *impact*
- **Risks and failures** can be mitigated through security measures, backup systems, disaster recovery plans, and monitoring tools. *mitigate*
- Examples of risks and failures include security breaches, data loss, system downtime, performance issues, and compliance violations.

incidents



1



2



3



4

observability
real-time
alerts



Risks and Failures in Information

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not
scale

all
Capacity
Crash



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[Sub-system]
- information system
expert system



Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.

- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.

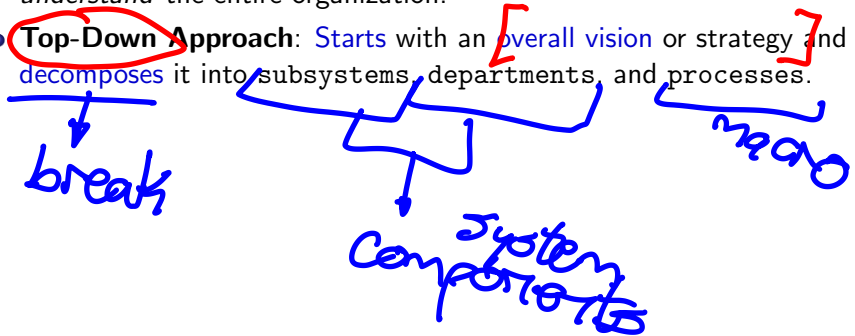
persons
tasks

BPMN
Context



Enterprises: Bottom-Up and Top-Down Approaches

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PIECE Framework for Enterprises

- **Participation** (*Engaging stakeholders*) at every level.
- **Independence of Thought**: Encouraging *diverse, innovative ideas*.
- **Elaboration**: Developing and *refining ideas and processes*.
- **Communication**: Ensuring *clear, effective exchange* of information.
- **Exploration**: Embracing *continuous innovation and improvement*.

Steve Jobs

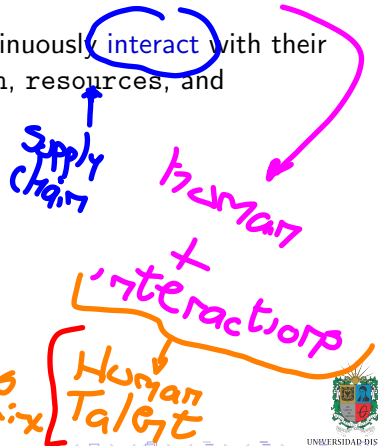
Concepts

strategy
SCRM
BPMN



Enterprise System Typologies

- **Rational Systems:** Organizations driven by logical structured processes and clear hierarchies.
- **Natural Systems:** Organizations viewed as self-organizing entities with emergent behavior.
- **Open Systems:** Enterprises that continuously interact with their external environment for information, resources, and innovation.



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies. → Data Analysis
- **Examples:** ERP systems, CRM systems, SCM systems.

• **Business Models:** Describe how an organization creates, captures value.

• Examples include subscription-based, freemium, pay-per-use, pay-per-form-based, and direct sales models.

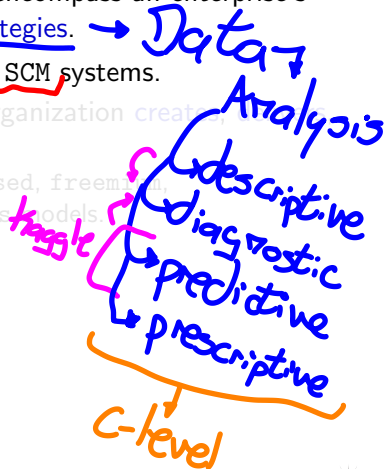
ERP

Enterprise Resource
Planning

CRM

Customer Relationship
Manager

SCM Supply Chain Management



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies.
- **Examples:** ERP systems, CRM systems, SCM systems.
- **Business Models:** Describe how an organization creates, delivers, and captures value.
 - Examples include subscription-based, freemium, platform-based, and direct sales models.

Netflix

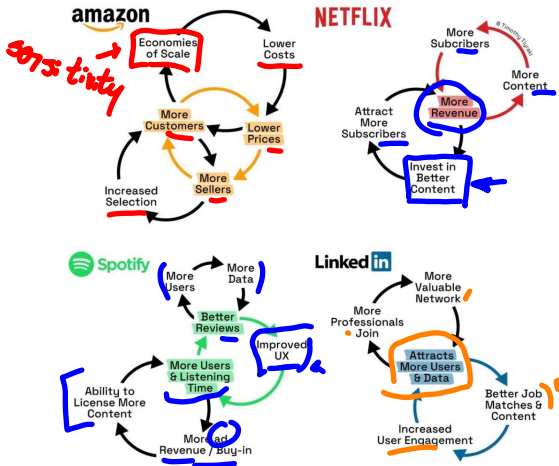
Spotify

Factory



Business Models Examples

Understanding Business Models Through Flywheels



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Basic Concepts in Software Methodologies

- **Software methodologies** provide frameworks for planning, designing, developing, testing, and maintaining software projects.
- They help teams manage project complexity and ensure quality deliverables.

feasible



Traditional Methodologies ~ 1960-1990

- **Waterfall:** A linear approach where each phase must be completed before moving to the next.
- Suitable for projects with well-defined requirements and low uncertainty.
- Emphasizes thorough documentation and planning.

Requirements

Analysis & Design

Implementation

Testing

Deploy & Maintaining



Traditional Methodologies ~ 1970-1985

- **Waterfall:** A linear approach where each phase must be completed before moving to the next.
- Suitable for projects with [well-defined requirements] and low uncertainty.
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20-30 documents

→ rigid

- Big companies - specific small project
↳ maths



Agile Methodologies — 1990s < internet startups

- Emphasize iterative development, customer collaboration, and flexibility.
- Based on the Agile Manifesto, which values individuals and interactions over processes and tools.

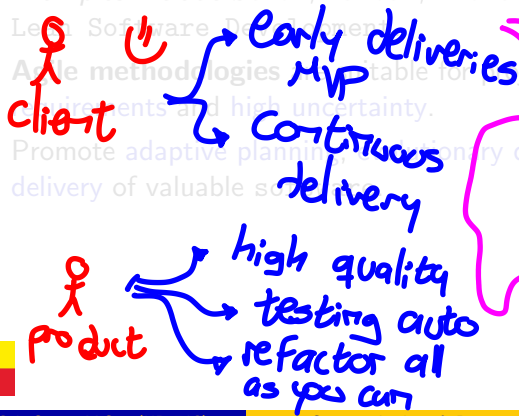
Examples include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.

Agile methodologies are suitable for projects with rapidly changing requirements and high uncertainty.

Promote adaptive planning, evolutionary development and early delivery of valuable software.

12 items

— Communication
— self-organized
— continuous improvement



Agile Methodologies

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start-up

rigid

Pair Programming
on-boarding



Agile Methodologies

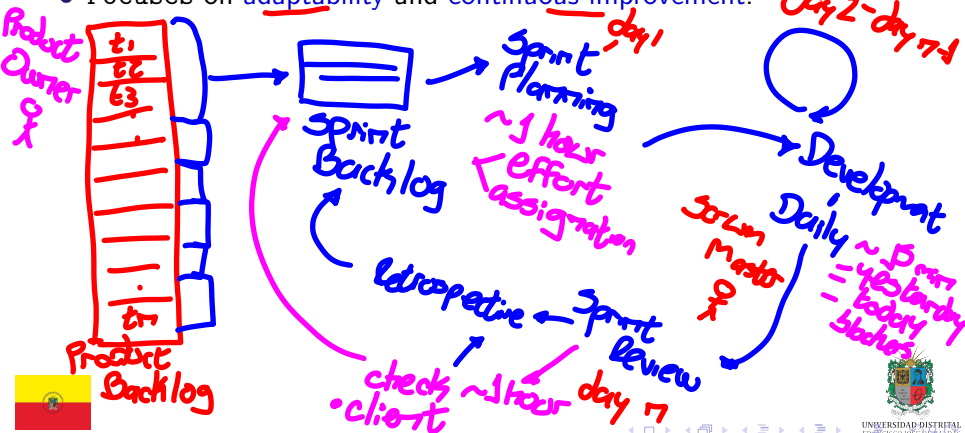
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Case Study: Scrum

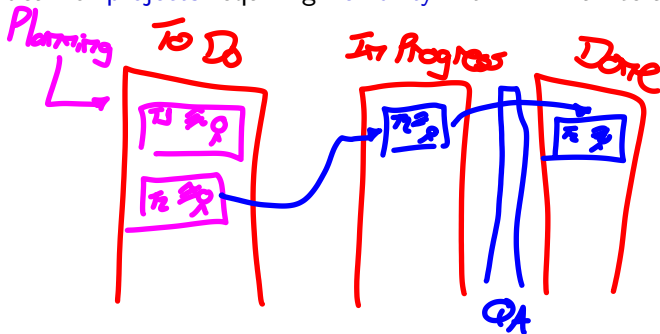
2 weeks ~ 1 week

- **Scrum** employs short, time-boxed iterations called sprints.
- Key practices include daily stand-ups, sprint planning, reviews, and retrospectives. → **ceremonies**
- Focuses on adaptability and continuous improvement.



Case Study: Kanban

- **Kanban** visualizes work items on **boards** and limits Work In Progress (WIP).
- Emphasizes gradual improvements, flow management, and continuous delivery.
- Ideal for **projects** requiring **flexibility** with *minimal iteration planning*.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

